Chiral Condensate Analysis in the Bosonized Lattice Schwinger Model Using the Hamiltonian Monte Carlo Algorithm

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Abstract

The Schwinger model provides a useful theoretical framework for studying non-perturbative phenomena relevant to quantum chromodynamics (QCD), including chiral symmetry breaking and topological effects. In this work, we investigate the temperature and vacuum-angle ($\boldsymbol{\theta}$) dependence of the chiral condensate in the bosonized lattice Schwinger model using the Hamiltonian Monte Carlo (HMC) algorithm. The chiral condensate $\langle\overline{\psi}\psi\rangle$ is evaluated for the massless and massive models at finite temperature and for nonzero values of $\boldsymbol{\theta}$. The results show that the chiral condensate decreases with increasing temperature and approaches the behavior expected for restored chiral symmetry at sufficiently high temperatures. The dependence on the vacuum angle exhibits a nontrivial variation of the condensate, with the observed behavior being consistent with the possibility of spontaneous CP symmetry breaking in the appropriate parameter regime. The HMC simulations also show increased autocorrelation and statistical uncertainty as the lattice spacing is reduced, indicating a slowing down of the sampling process. The inclusion of topological effects further increases autocorrelation times, indicating topological slowing down in the simulations. Overall, the simulation reproduces the qualitative behavior reported in previous studies and demonstrates the effectiveness of HMC for investigating thermodynamic and topological properties of the bosonized lattice Schwinger model.

Keywords: Schwinger Model, Hamiltonian Monte Carlo, HMC, Chiral Condensate

1 Introduction

Following the big bang, the universe underwent two types of phase transitions that are closely linked to mass generation in elementary particles, namely the electroweak phase transition and the quantum chromodynamics (QCD) phase transition [1]. The electroweak phase transition introduces the phenomenon of spontaneous symmetry breaking and triggers the emergence of the Higgs mechanism which generates the mass of various elementary particles such as fermions and bosons [2]. However, the contribution of the Higgs field to hadron mass is relatively small. Most of the hadron mass emerges during the QCD phase transition due to the phenomenon of spontaneous chiral symmetry breaking of QCD [3]. At low energy scales, QCD undergoes spontaneous chiral symmetry breaking which produces a chiral condensate and generates dynamic mass in hadrons [4, 5].

The Schwinger model is quantum electrodynamics (QED) in one spatial dimension introduced to demonstrate that a gauge field can acquire mass dynamically through quantum corrections that appear in the form of a chiral anomaly. This model shows a unique phenomenon where gauge invariance, which classically forbids the existence of an explicit mass term, can be violated by the chiral anomaly [6, 7]. This phenomenon gives rise to various physical properties such as the mass gap, and a chiral condensate which signifies the occurrence of chiral symmetry breaking [8]. Therefore, the Schwinger model can be used as a starting point to study more complex non-perturbative effects in gauge theory.

Despite being abelian and low-dimensional, the Schwinger model has similarities in various low-energy phenomena of QCD. Some of these similarities include chiral symmetry breaking, mass gap, confinement, and the topological θ-vacuum. Chiral symmetry breaking is an event that gives rise to dynamic mass acquisition in fermions, where in the case of the Schwinger model, this dynamic mass acquisition is manifested in bosons [9, 10]. The complex form and high numerical cost of running QCD simulations make the Schwinger model an ideal choice as a toy model for studying low-energy QCD phenomena. Because it possesses fewer degrees of freedom and requires lower computational costs, we can simulate the Schwinger model locally, utilizing it as a foundational step to gain a deeper insight into strong interactions [11].

Several cases within the Schwinger model have been extensively studied, and analytical solutions have been established for some of them. For instance, Sachs and Wipf [12] successfully derived the analytical solution for the chiral condensate in the massless Schwinger model at finite temperature. However, an analytical solution for the chiral condensate in the massive Schwinger model, including its dependence on the topological term (θ-vacuum), remains unavailable to date. This limitation has led us to rely on numerical methods, such as Markov Chain Monte Carlo (MCMC) and tensor networks, as alternative approaches to resolve this problem [10, 13].

Numerical solutions for quantum field theories, such as QED and QCD, generally necessitate a formulation on the lattice via spacetime discretization. The most prevalent framework in lattice field theory is Lattice Gauge Theory. However, Lattice Gauge Theory demands significant computational resources and gives rise to the notorious fermion doubling problem [10]. Given that the Schwinger model is formulated in 1+1 dimensions, we can implement bosonization to map the interacting fermion system

into a free bosonic theory [14]. This bosonization approach successfully reduces the computational overhead, bypasses the fermion doubling problem, and remains entirely free from the sign problem [10].

Hamiltonian Monte Carlo (HMC) is one of the most widely utilized methods for addressing various problems in quantum field theory, especially QCD [15]. This algorithm combines the Metropolis accept/reject scheme with Hamiltonian dynamics to generate field configurations that follow a specific probability distribution. HMC excels at exploring the sample space and due to its global proposals, it reduces autocorrelations between successive samples more efficiently than traditional Monte Carlo methods [16]. These advantages render HMC highly suitable for the bosonized lattice Schwinger model, in which we can numerically compute the fields by sampling configurations from a probability distribution governed by the system's action.

In this study, we develop a simulation program for the Schwinger model to extract the chiral condensate observable $\langle\overline{\psi}\psi\rangle$ utilizing the HMC algorithm. We select the chiral condensate $\langle\overline{\psi}\psi\rangle$ to investigate analogous QCD phenomena within a simplified framework, thereby validating our model before extending it to more complex systems. Furthermore, we utilize this program to investigate the behavior of the chiral condensate $\langle\overline{\psi}\psi\rangle$ as a function of temperature, as well as its dependence on the topological term (θ-vacuum). This approach allows us to analyze how the massive and massless variants, alongside the topological term, respond to thermal variations. Additionally, we investigate the underlying symmetry properties of the Schwinger model.

2 Bosonized Lattice Schwinger Model

Quantum electrodynamics (QED) is a quantum field theory that describes the interactions between matter and light (photons). It maps classical electrodynamics, governed by Maxwell's equations, into a fully quantized framework [2]. Mathematically, QED is formulated as an abelian gauge theory possessing a $U(1)$ symmetry group. Crucially, QED is invariant under local $U(1)$ gauge transformations; according to Noether's theorem, this local gauge invariance guarantees the conservation of electric charge. The Schwinger model is defined explicitly as the 1+1-dimensional analogue of QED. In Minkowski spacetime, the QED Lagrangian density is given by [17, 18]:

$$\mathcal{L}_{QED} = \overline{\psi} i\gamma^{\mu} \left[\partial_{\mu} + ig\,\mathrm{A}_{\mu}\right]\psi - m\overline{\psi}\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{1}$$

Where A_{μ} denotes the gauge field, ψ and its Dirac conjugate represent the fermion fields, g is the gauge coupling constant, γ^{μ} is the Dirac gamma matrices, m is the explicit fermion mass, and $F^{\mu\nu}$ represents the electromagnetic field strength tensor associated with the $U(1)$ gauge field.

Coleman [14] demonstrated that the angular parameter θ, known as the vacuum angle, can be introduced into the Schwinger model as a topological term within the Lagrangian. This term represents a background electric field capable of altering the vacuum structure of the system. One of the phenomena that occurs due to the influence of the topological term is Charge-parity (CP) violation. Unlike other terms in the Schwinger Lagrangian, this topological contribution is completely independent of the metric tensor $g_{\mu\nu}$; instead, it relies strictly on the totally antisymmetric Levi-Civita

symbol $\epsilon^{\mu\nu}$. Incorporating this topological term, the full Schwinger model Lagrangian in Minkowski spacetime is given by [2, 19]:

$$\mathcal{L} = \overline{\psi} i\gamma^{\mu} \left[\partial_{\mu} + igA_{\mu}\right] \psi - m\overline{\psi}\psi - \frac{1}{4} F_{\mu\nu}F^{\mu\nu} + \theta \frac{g}{4\pi} \epsilon^{\mu\nu} F_{\mu\nu}. \tag{2}$$

The model that will be used in this study is the bosonized lattice Schwinger model. This approach maps the fermion fields onto scalar fields, thereby bypassing both the doubling problem inherent in traditional Lattice Gauge Theory and the sign problem that appears the topological term following a Wick rotation. In this study, the Minkowski metric signature ($\eta_{\mu\nu}$) follows the $diag(+1, -1)$ convention. The formulated action of the bosonized Schwinger model is expressed as:

$$S_E = \int dx^2 \left[\frac{1}{2} \left(\partial_{\mu}\phi\right)^2 + \frac{g^2}{2\pi} \left(\phi + \frac{\theta}{2\sqrt{\pi}} \right)^2 - \frac{e^{\gamma}}{2\sqrt{\pi^3}} gm \, N_{g/\sqrt{\pi}} \cos\left(2\sqrt{\pi}\phi\right) \right]. \tag{3}$$

It can be seen that the action consists of three terms, namely the kinetic term, the effective photon mass term (which explicitly couples to the topological term), and the fermion mass term. The fermion mass term contains a normal-ordered product, denoted as ($N_{g/\sqrt{\pi}}$). This normal ordering can be eliminated by invoking Wick's theorem [20]. Specifically, Ohata [10] demonstrated that a UV-divergence factor ($\mathcal{O}(1/ag)$) can be systematically deployed to remove the normal ordering ($N_{g/\sqrt{\pi}}$). This elimination is essential for numerical tractability, as it transforms non-computable quantum operators into standard scalar fields. Subsequently, we discretize the resulting continuum action utilizing the finite-difference method and Riemann summation with lattice spacing a. The final expressions for the bosonized lattice Schwinger model action employed in our simulations are given by:

$$\begin{aligned} S_E = \sum_{\tau=0}^{L_{\tau}-1} \sum_{x=0}^{L_x-1} \Bigg[& \frac{1}{2}(\phi_{x+1,\tau} - \phi_{x,\tau})^2 + \frac{1}{2}(\phi_{x,\tau+1} - \phi_{x,\tau})^2 \\ & + \frac{(ag)^2}{2\pi} \left(\phi_{x,\tau} + \frac{\theta}{2\sqrt{\pi}} \right)^2 - \frac{e^{\gamma}}{2\sqrt{\pi^3}} \frac{m}{g} (ag)^2 \mathcal{O}(1/ag) \cos\left(2\sqrt{\pi}\phi_{x,\tau}\right) \Bigg]. \end{aligned} \tag{4}$$

The primary observable measured in this simulation is the chiral condensate $\langle\overline{\psi}\psi\rangle$, which serves as an order parameter to gauge the degree of chiral symmetry breaking. A vanishing chiral condensate ($\langle\overline{\psi}\psi\rangle = 0$) indicates that the system has undergone chiral symmetry restoration, signifying a phase where the chiral symmetry remains intact. Conversely, a non-zero expectation value implies the occurrence of spontaneous chiral symmetry breaking. Within the framework of the bosonized lattice Schwinger model, the formulation for the chiral condensate $\langle\overline{\psi}\psi\rangle$ is given by [10]:

$$\langle\overline{\psi}\psi\rangle = -\frac{e^{\gamma}}{2\sqrt{\pi^3}} g\mathcal{O}\left(1/ag\right) \left\langle \cos\left(2\sqrt{\pi}\phi_{x,\tau}\right) \right\rangle \tag{5}$$

3 Hamiltonian Monte Carlo

In this simulation, we employ the Hamiltonian Monte Carlo (HMC) algorithm. This approach introduces a fictitious momentum Π conjugate to the scalar field, thereby redefining the system's probability distribution in terms of a generalized Hamiltonian. The resulting canonical distribution within the phase space spanned by the scalar field ϕ and its momentum Π is defined as:

$$P\left(\phi, \Pi\right) \propto \exp\{(-H\left(\phi, \Pi\right))\}. \tag{6}$$

where $H(\phi, \Pi)$ is the Hamiltonian function that represents the total energy and follows the structure of a physical Hamiltonian system, $H(\phi, \Pi) = U(\phi) + K(\Pi)$. $K(\Pi)$ is the kinetic energy that arises due to fictitious momentum, while $U(\phi)$ corresponds to the potential energy determined by the lattice action. The Hamiltonian dynamics governing the system dictate the evolution of both the scalar field ϕ and its conjugate momentum Π. The Hamiltonian dynamics are written as [11, 21]:

$$H\left(\phi, \Pi\right) = S_E\left(\phi_{x,\tau}\right) + \sum_{x,\tau} \frac{\Pi_{x,\tau}^2}{2}. \tag{7}$$

The HMC algorithm generates momentum samples, which are subsequently utilized to update the configuration of the scalar field via the leapfrog integration scheme. The leapfrog integrator requires the gradient of the lattice action with respect to the field, a quantity formally referred to as the molecular dynamics force or just force in this case. This gradient is explicitly defined in Eq. (8).

$$\begin{aligned} F &= -\frac{\partial S_E}{\partial \phi_{x,\tau}} \\ &= -4\phi_{x,\tau} - (\phi_{x+1,\tau} + \phi_{x-1,\tau} + \phi_{x,\tau+1} + \phi_{x,\tau-1}) \\ &\quad + \frac{(ag)^2}{\pi}\left(\phi_{x,\tau} + \frac{\theta}{2\sqrt{\pi}}\right) + e^{\gamma}\frac{(ag)^2}{\pi}\frac{m}{g}\mathcal{O}(1/ag)\sin\left(2\sqrt{\pi}\phi_{x,\tau}\right). \end{aligned} \tag{8}$$

The leapfrog method is the most widely adopted numerical integration scheme in HMC simulations. This approach is necessitated by the fact that the underlying Hamiltonian dynamics cannot be solved analytically, requiring an approximation via numerical integration instead. As demonstrated by Brooks *et al.* [16], the leapfrog integrator exhibits stability for oscillatory motion, provided that the trajectory time step remains constant. Within a single integration step, the algorithm computes the updated approximations for both the scalar field and its conjugate momentum at the subsequent time interval $\tau + \epsilon$.

Upon completion of the leapfrog trajectory, where the updated values of the scalar field and momentum are determined, a Metropolis filter is applied to accept or reject the newly proposed state based on the change in the Hamiltonian energy. The field and momentum configurations from both the current and proposed states are substituted back into the Hamiltonian function. The proposed configuration is then accepted as the next state in the Markov chain with an acceptance probability given by:

$$P_{\text{acc}}\left(\phi_{x,\tau}, \phi^*_{x,\tau}\right) = \min\left[1,\, \exp\Big(H(\phi_{x,\tau}, \Pi_{x,\tau}) - H(\phi^*_{x,\tau}, \Pi^*_{x,\tau})\Big)\right]. \tag{9}$$

Here, $(\phi_{x,\tau}, \Pi_{x,\tau})$ is the current state and $(\phi^*_{x,\tau}, \Pi^*_{x,\tau})$ is the proposed state. If the transition is accepted, then the scalar field (ϕ) from the proposed state will become the new input in the next iteration of the Markov Chain. The ideal acceptance probability (P_{acc}) to be obtained is around 60%-80%. To achieve this optimal range, we dynamically adjust the integration step size ϵ via an automated calibration procedure. This calibration routine executes short HMC trial runs with a low number of configurations across a spectrum of step sizes ϵ until the acceptance rate stabilizes around 70%. Due to parameter variations, each simulated physical point requires a distinct optimal step size ϵ. Consequently, only the specific step size ϵ that satisfies this target acceptance criteria is deployed in the production simulation.

We execute the simulation in two distinct phases: thermalization and production. The thermalization phase is conducted to ensure that the system successfully reaches thermal equilibrium and is free from the bias of the initial configurations. This is achieved by discarding the early portion of the generated trajectories. Following the completion of the thermalization phase, we proceed to the production stage to sample the configurations of both the scalar field and its conjugate momentum. Specifically, the thermalization stage consists of 5×10^3 configurations, whereas the production stage comprises 5×10^4 configurations. The production-grade scalar field configurations ϕ are subsequently utilized to evaluate the target observable, namely the chiral condensate $\langle\overline{\psi}\psi\rangle$.

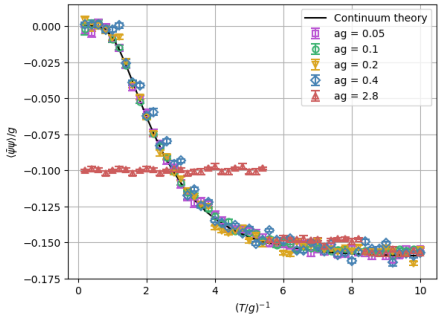
The spatial lattice size used in this work scales with the dimensionless parameter (ag), which is $L_x ag = 22.4$. We analyzed the temperature dependence through variations of the dimensionless parameter $(T/g)^{-1}$, which is directly related to the temporal lattice size $(T/g)^{-1} = L_\tau ag$ [10]. To evaluate the computational efficiency of the simulation, autocorrelation calculations are performed through the integrated autocorrelation time (τ_{int}) using Sokal's formulation [22]. statistical error bars for the simulated chiral condensate $\langle\overline{\psi}\psi\rangle$ are estimated using the Jackknife resampling method.

4 Results

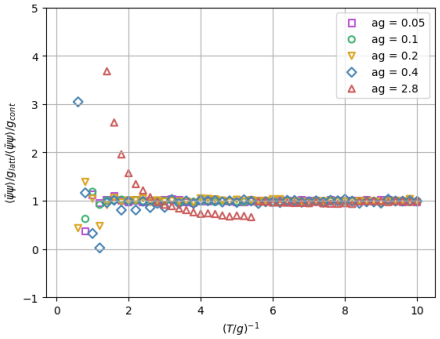
4.1 Massless Schwinger Model

Based on the data presented in Fig. 1a, the simulation successfully replicates the analytical trends within the lattice spacing range of $ag = 0.05$ to 0.4. Conversely, at a coarser lattice spacing of $ag = 2.8$, the temporal lattice size L_τ remains identical across several distinct temperatures, which prevents an optimal replication of the analytical curve. As the temperature increases, the chiral condensate systematically approaches zero, confirming the occurrence of chiral symmetry restoration. An interesting thing occurs at low temperatures, where the graph shows a converging trend around the

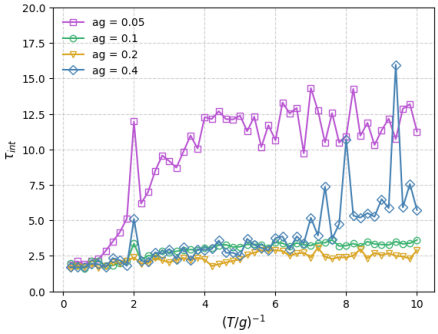
value of $-0.15g$. This convergence behavior has been shown by Sachs and Wipf [12], where at low temperatures ($T \approx 0$) the exact value of the condensate is $-0.1599g$.



(a)



(b)



(c)

Fig. 1: (a) Chiral Condensate as a function of temperature in the massless Schwinger model. (b) Ratio with respect to the continuum prediction. (c) Integrated autocorrelation time.

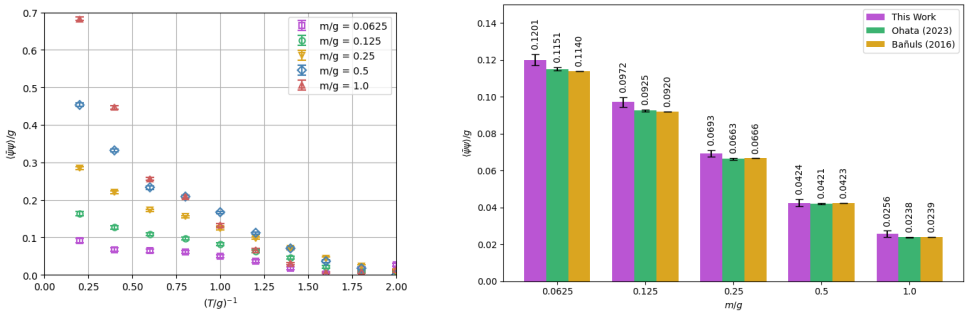
As illustrated in Fig. 1b, which depicts the ratio between the simulation results and the analytical theory, a significant divergence emerges in the high-temperature regime. These specific data points correspond to the region where chiral symmetry restoration takes place. The deviation in the ratio at elevated temperatures is reaching up to eight orders of magnitude at its peak. As the temperature decreases, this deviation reduces and the simulation results converge toward the analytical prediction. Consequently, the ratio exhibits a stable, convergent trend at lower temperatures. This indicates that while the absolute values of the condensate exhibit relatively small deviations at high temperatures, their corresponding ratios can amplify these differences, leading to a substantial order-of-magnitude discrepancy and extreme divergence.

To evaluate the statistical reliability of our simulations, we compute the integrated autocorrelation time τ_{int}. As illustrated in Fig. 1c, τ_{int} exhibits a tendency to increase

with decreasing temperature. Notably, the finest lattice spacing ($ag = 0.05$) displays a substantially elevated autocorrelation time. This high autocorrelation renders the observables at this specific spacing more prone to fluctuations and larger deviations compared to those at a coarser spacing, such as $ag = 0.1$. The inflation of τ_{int} directly leads to a reduction in the number of statistically independent samples. This behavior shown a notorious bottleneck in lattice field theory simulations known as the slowing down phenomenon. As the system approaches the continuum limit ($ag \to 0$), the correlation length diverges, which forces the HMC algorithm to explore the configuration space much more extensively to yield uncorrelated samples. Consequently, this surge in autocorrelation confirms the onset of slowing down near the continuum, which degrades the statistical reliability and reduces the computational efficiency of the simulation [23].

4.2 Massive Schwinger Model

The analytical solution of the massive Schwinger model has not been found, so numerical methods have become very popular for this problem. The explicit fermion mass term is well known to drive chiral symmetry breaking. In this work, we perform simulations to explore the behavior of the chiral condensate $\langle\overline{\psi}\psi\rangle$, focusing particularly on the low-temperature regime. Our numerical results are subsequently benchmarked against established reference data from prior studies utilizing MCMC heatbath and tensor network methods [10, 24]. The simulations are carried out at a fixed lattice spacing of $ag = 0.1$. To isolate the UV-finite component of the chiral condensate $\langle\overline{\psi}\psi\rangle$, we use a subtraction scheme where the free chiral condensate $\langle\overline{\psi}\psi\rangle_{\text{free}}$ is subtracted from the interacting massive condensate $\langle\overline{\psi}\psi\rangle_m$. Here, the free chiral condensate represents the non-interacting limit of the system, formulated in the absence of gauge coupling ($g = 0$).



(a) (b)

Fig. 2: (a) Chiral condensate as a function of temperature in the massive Schwinger model. (b) Comparison of the HMC results obtained in this work against the reference curves computed via the heatbath and tensor network frameworks from Refs. [10, 24].

The simulation results obtained show the trend of each mass variation. Fig. 2a shows how mass increases the value of the condensate at high temperatures. As the temperature of the system decreases, each mass variation experiences a decrease in the condensate. Although the trend shows a decrease in the condensate, no true chiral symmetry restoration occurs even at low temperatures. This trend at high temperatures is similar to the results obtained by Ohata [10]. Explicitly, the mass term has broken the chiral symmetry so that symmetry restoration cannot occur. The observed decrease in the condensate value merely implies a partial restoration. Consequently, the system does not undergo a distinct phase transition that would otherwise signify a full restoration of the chiral symmetry.

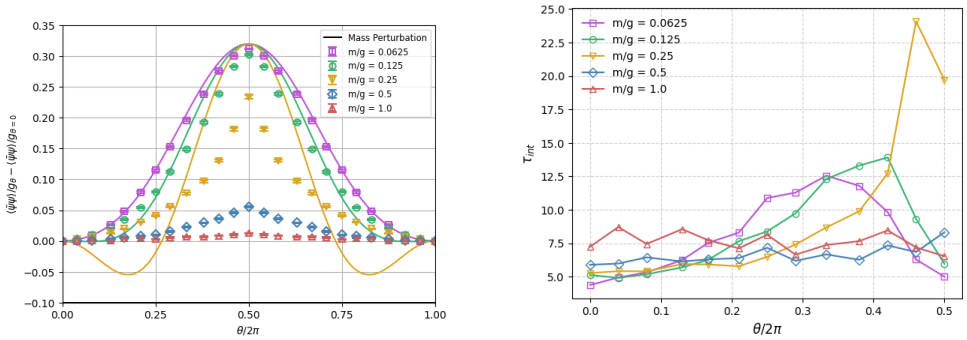
A comparison between our simulation results and the reference data at the low-temperature condition ($(T/g)^{-1} = 11.2$) is presented in Fig. 2b. All three datasets exhibit an identical qualitative trend, where the chiral condensate systematically decreases as a function of increasing fermion mass. However, the condensate values obtained in this work display a noticeable deviation from both benchmarks, with the maximum discrepancy reaching up to 7%. At lower mass regimes, this relatively high deviation diminishes the statistical significance of our results, as the data points fall outside the established uncertainty bounds. Conversely, in the heavy-mass regime ($m/g = 0.5$ and $m/g = 1.0$), our simulation results remain fully consistent with the reference data within the estimated error bars.

The discrepancy observed in the lower-mass regime warrants a closer examination, suggesting two plausible underlying causes. The first possibility stems from an insufficient ensemble of independent samples, dictated by the limited number of configurations generated. Specifically, the integrated autocorrelation time spans approximately $3-4\ \tau_{int}$, yielding roughly $6,000-8,000$ statistically independent samples. While enhancing the statistics via a larger configuration size would naturally improve precision and suppress error bars, this approach was unfeasible in the present study due to lack of computational resource. The second possibility arises from finite-temperature effects. Since our simulation is performed at an inverse temperature of $(T/g)^{-1} = 11.2$, a non-vanishing thermal contribution remains present in the system, albeit small. This fundamentally contrasts with the zero-temperature continuum interpolation reported by Bañuls [24] using the tensor network framework.

4.3 Schwinger Model with Topological Term

This section investigates the behavior of the chiral condensate $\langle\overline{\psi}\psi\rangle$ acting as representation for chiral symmetry in response to variations in the vacuum angle θ introduced by the topological term. Similar to the massive case, an analytical solution for this topological formulation remains unavailable to date, thereby necessitating the application of numerical methods. The simulations are performed at a fixed lattice spacing of $ag = 0.1$ under a constant temperature corresponding to an inverse temperature of $(T/g)^{-1} = 11.2$. The numerical results obtained herein are subsequently evaluated against the leading-order mass perturbation theory and reference data from MCMC heatbath simulations [10, 25].

The results presented in Fig. 3a reveal a characteristic cusp-like behavior in the condensate profile. Both our simulation results and the reference data demonstrate that



(a) (b)

Fig. 3: (a) Chiral condensate in the Schwinger model with the presence of the topological term. (b) Integrated autocorrelation time.

an increase in the fermion mass systematically suppresses the magnitude of the condensate, which is evident from the attenuation of the peaks as the mass increases [10]. However, this behavior qualitatively deviates from the predictions of the leading-order mass perturbation theory (represented by the solid lines), where the maximum peak heights remain constant and invariant under mass variations. Consequently, this discrepancy restricts the applicability of the leading-order mass perturbation framework strictly to the light-mass regime.

The statistical reliability of the simulation can be seen through the obtained autocorrelation. It can be seen from Fig. 3b where there is a significant increase in the autocorrelation effect at a mass of $m/g = 0.25$. An approximation value of 20 for τ_{int} is obtained at the peak condition. An autocorrelation of this magnitude indicates a less efficient simulation. Compared to the massive and massless models, the autocorrelation obtained in this case is quite significant with $\tau_{int} > 5$. This significant increase causes a decrease to 1000 independent samples, indicating that 98% of the samples are correlated with each other. This problem is another form of slowing down that arises due to the presence of the topological dependence of the vacuum angle θ. The same case also occurs in QCD and is known as topological slowing down [26].

The obtained numerical results clearly unveil a sharp cusp-like profile in the condensate, which becomes particularly pronounced at larger fermion masses of $m/g = 0.5$ and 1.0. Ohata [10] conjectured that this specific behavior potentially signals the occurrence of spontaneous CP symmetry breaking. Such a phenomenon, which indicates the violation of CP symmetry, similarly emerges in Quantum Chromodynamics (QCD) at the topological vacuum angle $\theta = \pi$ [27]. Typically, this symmetry-breaking event is restricted to the zero-temperature limit under sufficiently heavy fermion masses, which aligns qualitatively with our observations. A definitive confirmation remains precluded due to the non-vanishing thermal contributions stemming from the finite-temperature constraints of the present simulation.

5 Conclusion

HMC (Hamiltonian Monte Carlo) simulation has been performed to find the value of the chiral condensate $\langle\overline{\psi}\psi\rangle$ in three types of Schwinger models, namely massless, massive without a topological term, and massive with a topological term. The success of this simulation proves that HMC is a valid method used to solve problems in the bosonized Schwinger model. The results obtained from the simulation have also been compared and validated using other results in the literature. Although only using 5×10^4 configurations, there is no significant difference between the simulation results and the reference except for some small discrepancies, with all results showing the same trend as the reference results.

For the massless Schwinger model, temperature changes can affect the chiral symmetry to undergo restoration at high temperatures, and conversely break the symmetry at low temperatures. The massive model without a topological term shows how the condensate experiences a decrease as the temperature decreases, but no chiral symmetry restoration occurs due to the presence of the mass term that explicitly breaks the chiral symmetry. In the massive model with a topological term, the condensate has a cups like trend with the condensate peak located at the vacuum angle $\theta = \pi$. The shape of the graph indicates the possible emergence of spontaneous CP symmetry breaking which can break the CP (Charge-parity) symmetry.

Declarations

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Conflict of interest/Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Data availability

The data generated and analysed during this study are available from the corresponding author upon reasonable request.

Code availability

The code used to generate the results presented in this study is publicly available at https://github.com/putrapratama8789-wq/Bosonized_Lattice_Schwinger_Model_HMC.git.

Author contribution

The author conceived the study, developed and implemented the numerical methods, performed the simulations, analysed the results, and wrote the manuscript.

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